
When Shift Happens - Confounding Is to Blame

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Abstract

1 Distribution shifts are a pervasive obstacle across the many settings in which
2 machine learning models are deployed, introducing uncertainty that undermines
3 their robustness and generalization. While conventional wisdom holds that learning
4 causal-invariant representations enhances robustness to such shifts, a growing
5 body of empirical work reports a counterintuitive and broadly recurring finding:
6 (i) empirical risk minimization (ERM) can rival or even outperform state-of-the-
7 art out-of-distribution (OOD) generalization methods across diverse tasks and
8 model families, and (ii) its OOD generalization performance improves when all
9 available covariates—not just causal ones—are utilized. Drawing on both empirical
10 and theoretical evidence, we trace this phenomenon to hidden confounding, a
11 mechanism that arises broadly whenever data are gathered across heterogeneous
12 environments. Shifts in hidden confounding induce changes in data distributions
13 that violate assumptions commonly relied upon by existing OOD generalization
14 approaches. Under such conditions, we prove that effective generalization generally
15 requires learning environment-specific relationships, rather than relying solely on
16 invariant ones. We further show that models augmented with proxies for hidden
17 confounders can mitigate the challenges posed by hidden confounding shifts. Taken
18 together, these findings offer theoretical insights and practical guidance that apply
19 across a wide range of prediction problems, informing the design of robust OOD
20 generalization algorithms and principled covariate selection strategies.

1 Introduction

21 Generalization—the ability to draw reliable conclusions about unseen data based on observed data—is
22 a unifying goal across the sciences and engineering. It recurs under different names in fields that
23 otherwise share little: in medicine and the social sciences, it is embodied as *external validity*, ensuring
24 findings from one population are applicable to a different population [13]; in ecology, it supports
25 *space-for-time substitutions*, where spatial variation is used as a proxy for temporal change to infer
26 long-term ecological patterns [60]; and in engineering, it drives *robust control*, where models must
27 maintain performance in the presence of unmodeled disturbances [35]. The breadth of these instances
28 underscores that generalization is a shared concern wherever models are transported across conditions.
29 In recent years, machine learning has become a powerful and widely adopted tool for learning
30 generalizable models [100, 88, 49].

32 To generalize well, machine learning models must be robust to distribution shifts between training and
33 test data—a requirement that arises whenever deployment conditions depart from those seen during
34 training. For example, a model trained to predict *food stamp reciprocity* from *household attributes*
35 in one region should adapt and continue to perform well when deployed in another, and analogous
36 demands appear across countless prediction tasks. A model is said to achieve *out-of-distribution*
37 (*OOD*) *generalization* when it maintains its performance on both in-distribution (ID) data (from which
38 training data are sampled) and out-of-distribution (OOD) test data. Over the last decade, broad families
39 of methods—including domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98,
40 82, 51, 90, 91, 81], and robust learning [42, 69, 94]—have been proposed to achieve provable
41 OOD generalization under specific assumptions. Yet, under careful model selection [30, 38, 87],
42 models based on standard *empirical risk minimization* (ERM) [86] repeatedly attain competitive OOD

generalization performance across a wide spectrum of real-world applications [30, 24, 50, 57, 67, 87]. Moreover, a recent empirical study [57] spanning 16 real-world tabular datasets concludes that incorporating all available covariates when predicting the outcome—regardless of whether they directly affect it—can improve OOD generalization performance. Because these observations recur across many benchmarks and method families, they challenge prevailing assumptions in the field and motivate a deeper investigation into the general mechanisms underlying OOD generalization.

Distribution shifts arise generically whenever data originate from different environments [71]. For instance, *food stamp reciprocity* may differ across states because each operates under different eligibility rules, yielding distinct, environment-specific distributions—a pattern that has direct analogues across domains, populations, and time periods. In such data, certain statistical relationships between covariates and the outcome may remain consistent across all environments; these are referred to as *invariant relationships* [6]. Identifying and learning such invariant relationships has been a central route to models with OOD generalization [6, 59, 31, 66, 56, 98, 63]. A special case is *causal invariance*, in which the causal relationships in the data stay invariant across environments [71, 59, 6]. However, across a broad range of real-world settings, not all covariates relevant for predicting an outcome are observed, owing to limitations in data collection, privacy constraints, or measurement errors [14, 52, 20, 48]. Their absence disrupts the invariant relationships that models rely on for generalization. The difficulty is compounded when the unobserved variables are confounders that influence both the observed covariates and the outcome. Such hidden confounders are pervasive in practice, and shifts in their distributions correspond to distinct environments. Ignoring these shifts not only undermines generalization performance [40, 4, 85, 62], but can also lead models to learn incorrect relationships between observed covariates and the outcome—a failure mode that generalizes well beyond any single application.

Despite its importance across application areas, the impact of hidden confounding shifts on OOD generalization remains broadly underexplored [4, 40, 85, 62]. This work offers both theoretical insights and empirical justifications that together explain a range of recent findings in the literature [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- We motivate the need to study confounding shift in OOD generalization from a general causal perspective (§ 3) and explain why adding informative, non-causal covariates can improve performance across settings.
- More broadly, we show that maximizing the predictive information between model predictions and true outcomes fundamentally demands learning environment-specific relationships. Consequently, invariant representations alone, while necessary, are insufficient in general (§ 4.1, § 4.2).
- We demonstrate that variables informative of either the outcome or the hidden confounders help improve the predictive information between model predictions and true outcomes. This holds regardless of whether such variables are causal, underscoring the value of principled covariate selection whenever hidden confounding shift is present (§ 4.3).
- Our experiments on both real-world and synthetic datasets provide converging evidence that (i) hidden confounding is prevalent in real-world tabular benchmark data, (ii) learning environment-specific relationships correlates positively with ID and OOD test accuracy, and (iii) incorporating informative, non-causal covariates improves generalization (§ 5).

Taken together, our work not only explains the surprising effectiveness of ERM under real-world distribution shifts, but also brings out the subtle yet inevitable role that hidden confounding plays in OOD generalization more generally—highlighting the need for methods that address it more effectively across the many settings where it arises.

2 Related work

Out-of-distribution (OOD) generalization spans a wide array of problem formulations, with prominent examples including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. These settings differ in the information they assume: domain adaptation assumes access to unlabeled data from the test set, whereas domain generalization relies on environment labels during training; federated learning adopts a collaborative framework that tackles constraints such as communication efficiency and privacy when data originate from multiple environments; and OOD detection focuses on identifying samples that differ substantially from the training distribution.

Despite their differences, a goal shared across many of these methods is to learn invariant relationships. Recent work, however, indicates that inductive biases beyond invariance are broadly required for improved generalization [46, 72, 93]. Ye et al. [93] argue that invariance of features is necessary but not sufficient for generalization and highlight the importance of feature *informativeness*. We likewise study the roles of both invariance and informativeness, and our analysis applies across these otherwise distinct OOD generalization settings.

Proxy variable adjustment: When confounding variables are observed during training and unobserved at test time, Landeiro and Culotta [40] propose adjusting for confounding shifts to improve classifier performance. Building on proxy-based adjustment methods for causal effect estimation [55, 39], a line of domain adaptation methods recasts the problem of unknown distribution shifts as a causal effect identification problem [4, 85]. More recently, OOD generalization under hidden confounding shift has been studied under the assumption of overlapping confounder support [62]. In contrast to these approaches, we provide a general account of *how* proxy variables enhance OOD generalization, applicable beyond any specific adjustment procedure.

All variable models vs causal models. In a benchmark study spanning many prediction tasks, Nastl and Hardt [57] categorize covariates into four groups: causal (conservatively chosen), arguably causal, anti-causal, and other spurious covariates. Across 16 benchmark datasets, they show that models using all covariates Pareto-dominate those using only causal or arguably causal subsets on both ID and OOD data—a strikingly consistent pattern across the datasets considered. Theoretical explanations for this behavior remain scarce, and we develop scenarios and arguments that account for it. For linear causal models, anchor regression [68] introduces a framework that interpolates between two estimation paradigms: models that include all observed covariates and models that focus solely on causal covariates. We aim to explain, more generally, the impact of adding covariates that are not necessarily causal under a hidden confounding shift. Eastwood et al. [21] show that unstable covariates can boost performance when they carry information about the label, provided they are conditionally independent of the stable covariates given the label, and propose adjusting for the shift by examining the test domain without labels. Yet even when their method is applied to a medical real-world dataset not constructed for this particular problem [8], ERM remains competitive, echoing the findings of Nastl and Hardt [57] across settings. This is consistent with the broader insight that, under well-specified covariate shifts, maximum likelihood estimation (MLE) achieves minimax optimality for OOD generalization [26]. Real-world settings, however, are rarely well-specified, precisely because of hidden confounding shift—the general phenomenon that is the main focus of this work.

3 Manifestations of hidden confounding shift

In this section, we provide background on hidden confounding shift and motivate the need to address it, using an example involving causal effect identification of observed covariates on the outcome. The mechanisms we describe are not specific to any single dataset or task; they characterize distribution shifts broadly, whenever data are collected across differing environments.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , several distinct distribution shifts can arise between any two environments e and e' , as summarized in Table 1. In general, these shifts trace back to a shift in the distribution $\mathbb{P}(U)$ of an unobserved covariate U that causes $\mathbf{X}$, Y , or both (see Figure 1). A shift in $\mathbb{P}(U)$ propagates into shifts in the observed distributions over $\mathbf{X}$ and Y . For instance, we can write $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$. Thus, when $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$, we observe $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. Parallel arguments apply to the shifts of $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$, so that a single underlying shift in $\mathbb{P}(U)$ can manifest as any of the observed shift types.

Many existing methods for OOD generalization assume particular causal relationships among U , $\mathbf{X}$, Y that guarantee specific invariances, and each such assumption carves out a special case

Table 1: A summary of the different distribution shifts that can arise across environments.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

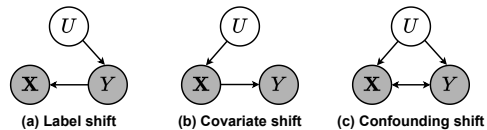


Figure 1: Causal graphs underlying the various distribution shifts that can arise across environments.

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

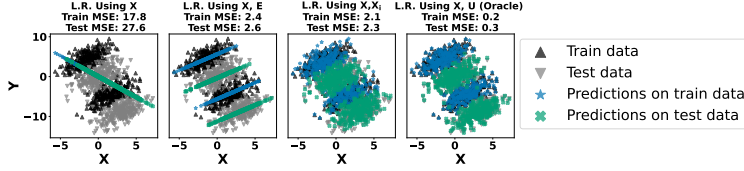


Figure 2: We evaluate four linear regression (L.R.) models in a representative OOD setting characterized by hidden confounding shifts and minimal environment overlap (i.e., distant μ_e). (i) A model trained solely on X learns an incorrect relationship with Y , an instance of Simpson’s paradox. (ii) Using environment-specific summary statistics of X , denoted E , recovers the correct relationship but remains limited in representational power. (iii) Using a covariate X_i informative for U improves OOD generalization. (iv) The oracle model is trained on X and U .

of the general picture. For instance, when $U \rightarrow Y \rightarrow X$ (Figure 1 (a)), since $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$, a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ arises through label shift, i.e., $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, while the conditional covariate distribution stays invariant, i.e., $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$, because $\mathbb{P}(\mathbf{X}|Y, U) = \mathbb{P}(\mathbf{X}|Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. Similarly, when $U \rightarrow X \rightarrow Y$ (Figure 1 (b)), the shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ arises through covariate shift, i.e., $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, while $\mathbb{P}(Y | \mathbf{X})$ stays invariant, i.e., $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. Such invariances fail to hold in the broad and practically common regime where U causes both $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. In that regime, a single shift in $\mathbb{P}(U)$ between environments can induce label shift, covariate shift, conditional covariate shift, and concept shift all at once. Even so, confounding shift preserves one invariance, $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$, referred to as *stable confounding shift* [85, 4]. Because U is unobserved, this invariance is not accessible from the observed data—a limitation that holds regardless of the specific domain.

Information-theoretic measures: Following Federici et al. [22], we quantify distribution shifts with mutual information, which provides a unified, model-agnostic vocabulary applicable to any covariate–target pair. Specifically, we use $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E|Y)$, $I(Y; E|\mathbf{X})$, $I(\mathbf{X}, Y; E)$, where E is the environment variable, to measure the shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$, respectively, across environments. As noted above, existing OOD generalization methods typically assume either $I(\mathbf{X}; E | Y) = 0$ through the label shift assumption or $I(Y; E | \mathbf{X}) = 0$ through the covariate shift assumption. In the general presence of hidden confounding shift, however, both quantities may be nonzero.

Challenges with hidden confounding shift: To illustrate the challenges that hidden confounding shift poses for OOD generalization, we present the representative experiment shown in Figure 2. The outcome Y depends on X and a hidden confounder U , and environment-specific shifts in the mean of U , denoted μ_e , induce distinct environments. A linear regression model trained solely on the observed covariate X infers an incorrect relationship between X and Y , as shown in the first subplot. This is an instance of the broadly documented Simpson’s paradox [73]: Y increases with X within each environment, yet the model captures the opposite trend. The same failure appears for models designed for OOD generalization, such as IRM [6], VREX [38], and Group DRO [69], indicating that it is not an artifact of any single estimator. In such scenarios, we show theoretically in § 4 that the generally optimal strategy is to learn environment-specific relationships, which enables recovery of the correct relationship between X and Y . This is illustrated in the second subplot, where environment-specific summary statistics of X —such as its mean, standard deviation, and quantiles—help uncover the true X – Y relationship. This mirrors the *backdoor adjustment* criterion in causal effect estimation [58], where environment-specific information acts as a proxy for adjusting for confounders. The environment-specific information encoded in observed covariates alone may nonetheless be insufficient, owing to limited representational capacity. As we show in § 4.3, additional informative covariates that serve as proxies for the unobserved confounders can further improve generalization, as shown in the third subplot. For comparison, we include an oracle model trained on X and U , which attains the best-possible mean squared error.

4 Impact of hidden confounding shifts on OOD generalization

In this section, we study theoretically how hidden confounding shifts affect various aspects of OOD generalization. The analysis is deliberately general: it makes no assumptions specific to a particular task, data modality, or model family. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ be the predicted label for an input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation function. $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ denotes entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ denotes mutual information. The risk of the predictor $(f \circ \phi)$ in an environment e is defined as: $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X},Y) \sim \mathbb{P}_{\mathbf{X},Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ where $\ell(\cdot, \cdot)$ is a loss function. The goal in OOD generalization is to learn a predictor $(f \circ \phi)$ that performs well on both ID and OOD data. To this end, various objectives have been considered in the literature. Robust optimization-based methods [7, 9, 19, 75, 69] aim to minimize the worst risk across all training environments: $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ where $\mathcal{E}_{tr}$ denotes the set of training environments. The variance risk extrapolation (VREX) [38] aims to minimize $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$ where β is a hyperparameter. Invariant risk minimization (IRM) [6] aims to minimize $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ with the constraint that f is a simultaneously optimal classifier across all environments. Empirical risk minimization (ERM) based methods simply pool data from all training environments and minimize the empirical risk on the pooled data [6, 38]. That is, ERM minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X},Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$ where $\mathcal{D}^e$ denotes observed data in an environment e .

To understand how hidden confounding shift affects traditional objective functions, we examine different aspects of maximizing the predictive information $I(Y; \hat{Y})$, where $\hat{Y} = (f \circ \phi)(\mathbf{X})$. Since $I(Y; \hat{Y})$ quantifies how informative the prediction $\hat{Y}$ is about the true label Y , it serves as a natural and broadly applicable objective across machine learning tasks. We begin by defining two key properties of $\phi(\mathbf{X})$ —*informativeness* and *invariance*—both of which are crucial for OOD generalization in general [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness measures the information $\phi(\mathbf{X})$ provides about Y within each environment. Minimizing CINF implies information loss, and maximizing CINF can be undesirable in some applications, such as algorithmic fairness, where $\phi(\mathbf{X})$ may exploit sensitive or biased information within environments and thereby produce unfair predictions. Although such biased representations can be avoided by minimizing $I(\phi(\mathbf{X}); E)$, doing so may reduce predictive performance, as we show in § 4.2. In the broad class of settings where no sensitive information is associated with E , we adopt the perspective that maximizing CINF can improve generalization, a view we substantiate in § 4.1.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$.*

Invariance quantifies how consistent the representation $\phi(\mathbf{X})$ is across different environments for a given label Y . Minimizing invariance can lead to overfitting by preserving environment-specific information in $\phi(\mathbf{X})$, whereas maximizing invariance helps eliminate environment-specific dependencies and promotes invariant learning. Extending the information-theoretic measures from § 3, we use $I(\phi(\mathbf{X}); E)$ and $I(Y; E | \phi(\mathbf{X}))$ to quantify *feature shift* and *concept shift*, respectively. Depending on the context, we use *concept shift* to denote either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$; these measures are defined for arbitrary representations and thus apply uniformly across models.

4.1 A general decomposition of predictive information

We begin with a few causal-graph preliminaries. A causal graph $\mathcal{G}$ consists of nodes representing random variables, and directed edges indicating direct causal influences between nodes. A *path* between two nodes X_i and X_j is a sequence of unique nodes connected by edges. A *directed path* from X_i to X_j with $i < j$ is one where all arrows point toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. In such a directed path, X_i is called the *parent* of X_{i+1} , X_{i+1} is the *child* of X_i , X_i an *ancestor* of X_j , and X_j is a *descendant* of X_i . Paths decompose into three fundamental structures: a

246 chain $X_i \rightarrow X_j \rightarrow X_k$, a fork $X_i \leftarrow X_j \rightarrow X_k$, and a collider $X_i \rightarrow X_j \leftarrow X_k$. In both chains and
 247 forks, X_i and X_k are marginally dependent yet become conditionally independent upon conditioning
 248 on the intermediate node X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k \mid X_j$. In a collider, X_i and X_k are
 249 marginally independent but become conditionally dependent when conditioning on X_j or any of its
 250 descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k \mid X_j$. A path between X_i and X_k is said to be *blocked* by
 251 a conditioning set $\mathcal{S}$ if and only if either (i) the path contains a chain or fork whose middle node lies
 252 in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to $\mathcal{S}$.
 253 If all paths from X_i to X_k are blocked by $\mathcal{S}$, then $X_i \perp\!\!\!\perp X_k \mid \mathcal{S}$. A path is open if it is not blocked.

254 Now consider the predictive information $I(Y; \hat{Y})$, where the
 255 predictions $\hat{Y} = (f \circ \phi)(\mathbf{X})$ are based on a learned representa-
 256 tion $\phi(\mathbf{X})$. We model the underlying causal relationships
 257 among $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$ and E as in Figure 3, a structure
 258 general enough to subsume both the causal ($\mathbf{X} \rightarrow Y$) and
 259 anti-causal ($Y \rightarrow \mathbf{X}$) regimes that arise across prediction
 260 tasks. Here, U is a hidden confounding variable. Given ei-
 261 ther $\mathbf{X}$ or $\phi(\mathbf{X})$, $\hat{Y}$ is redundant for reasoning about Y . That
 262 is, $\hat{Y} \perp\!\!\!\perp Y \mid \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y \mid \phi(\mathbf{X})$. E denotes an environment
 263 variable that captures shifts in $\mathbb{P}(U)$. That is, between any two environments, there is a shift in $\mathbb{P}(U)$.
 264 We first present a more general decomposition of predictive information without explicit consideration
 265 of how U influences $\mathbf{X}, Y$. A formal treatment of U 's influence on $\mathbf{X}$ and Y is presented in § 4.2.

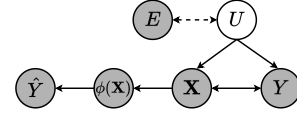


Figure 3: The bi-directed arrow between $\mathbf{X}$ and Y indicates the general case in which some covariates of $\mathbf{X}$ cause Y while others may be caused by Y .

266 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 267 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 268 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \underbrace{I(\phi(\mathbf{X}); Y|E)}_{\text{Cond. informativ.}} - \underbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}_{\text{Variation}} + \underbrace{\frac{I(Y; E)}{2}}_{\text{Label shift}} + \underbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}_{\text{Feature shift}} - \underbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}_{\text{Concept shift}} - \underbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}_{\text{Residual}} \quad (1)$$

269 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 270 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

271 Proofs are presented in Appendix § A. This decomposition offers a common lens through which a
 272 wide range of existing methods can be understood, each as implicitly minimizing or maximizing
 273 certain terms in Equation (1). For instance, IRM [6] enforces that a fixed classifier f remains
 274 the same across different environments. This is directly related to maximizing $-I(\phi(\mathbf{X}); E|Y)$
 275 because if $\phi(\mathbf{X})$ were to carry extra environment-specific information conditional on Y , the fixed
 276 classifier would no longer be optimal in all environments. DANN [23] and the independence
 277 criterion for fair classification [22] aims to minimize environment-specific information in $\phi(\mathbf{X})$
 278 by minimizing $I(\phi(\mathbf{X}); E)$. In the context of domain adaptation, certain properties of features
 279 are learned invariant to domains [82, 90]. While minimizing $I(\phi(\mathbf{X}); E)$ may be good for fair
 280 classification, it may degrade predictive performance [98, 33, 22]. This can be seen from Equation (1),
 281 where minimizing $I(\phi(\mathbf{X}); E)$ can reduce $I(Y; \hat{Y})$. In CDAN [51], the objective is to obtain
 282 $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$. By enforcing this equality, E becomes independent of the pair
 283 $(\phi(\mathbf{X}), Y)$. Consequently, $\mathbb{P}(Y \mid \phi(\mathbf{X}), E) = \mathbb{P}(Y \mid \phi(\mathbf{X})) \implies I(Y; E \mid \phi(\mathbf{X})) = 0$. Thus,
 284 although CDAN does not explicitly include $I(Y; E \mid \phi(\mathbf{X}))$ in its loss, its joint-distribution alignment
 285 objective effectively drives this conditional mutual information to zero. While $I(Y; E)$ is a constant,
 286 there exist methods that are robust to shifts in the distribution of labels [69, 95].

287 Recall that Equation (1) is derived without explicitly specifying how U influences $\mathbf{X}, Y$, and it
 288 therefore covers many causal configurations at once. Knowing the specific way in which U influences
 289 $\mathbf{X}$ and Y can further guide how the terms in (1) should be handled. If $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$,
 290 then $I(\mathbf{X}; E \mid Y) = 0$, so minimizing $I(\phi(\mathbf{X}); E \mid Y)$ is a principled objective. Similarly, if
 291 $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E \mid \mathbf{X}) = 0$, which motivates minimizing $I(Y; E \mid \phi(\mathbf{X}))$. In
 292 the more general and practically pervasive case where $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, the interaction among
 293 the terms in Equation (1) becomes non-trivial, and it is no longer obvious what constitutes an ideal
 294 strategy. We answer this question in the next section.

4.2 Predictive information under hidden confounding shift

To understand how a hidden confounding variable U that causes both $\mathbf{X}$ and Y affects the predictive information, we consider the *two canonical cases* of the relationship between $\mathbf{X}$ and Y : (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. Together these cover the full range of directionalities encountered across prediction tasks. When $\mathbf{X} \rightarrow Y$, we obtain the inequalities in (2) because conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider node Y , resulting in additional information flow from $\phi(\mathbf{X})$ to E via Y . Conversely, conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the node $\mathbf{X}$, as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Similarly, when $Y \rightarrow \mathbf{X}$, we obtain the inequalities in (3) because conditioning on Y blocks the information flow from $\phi(\mathbf{X})$ to E at the node Y and conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E at the node $\mathbf{X}$ because $\phi(\mathbf{X})$ is the child of $\mathbf{X}$ and conditioning on the child of a collider opens a path via that collider [58] (recall the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Using the inequalities in (2), (3), we refine the predictive information decomposition as follows.

Proposition 4.2. *For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

In Proposition 4.2, we treat the two cases $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, so that the result covers either directionality. The causal relationship $\mathbf{X} \rightarrow Y$ is the more prevalent one across real-world tabular prediction tasks: for instance, 11 of the 16 datasets considered in recent benchmark studies [57, 50, 24] follow the structure $\mathbf{X} \rightarrow Y$, and even in the remaining 5, the number of covariates caused by Y is far smaller than the number that cause Y . This suggests that the $\mathbf{X} \rightarrow Y$ case captures the dominant behavior in practice, while the proposition still accommodates the anti-causal case. We present the key takeaway from Proposition 4.2 below.

Compared to the decomposition in (1), the decomposition in (4) (which is obtained for the special cases: (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$, (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$) is free from the terms: *variation, label shift, feature shift, and concept shift*. That is, under hidden confounding shift, when either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, the predictive information can be maximized by maximizing the difference: *conditional informativeness* – *residual*.

Maximizing conditional informativeness: In recent work, Prashant et al. [62] proposed a mixture-of-experts (MoE) model for OOD generalization under hidden confounding shift, where each expert corresponds to a specific hidden confounder assignment. That is, each expert focuses on maximizing performance within the environment corresponding to a particular value of the hidden confounder. Learning MoE models therefore aligns with the goal of maximizing the conditional informativeness $I(\phi(\mathbf{X}); Y|E)$, and Equation (4) provides theoretical justification for MoE-type models. Beyond supporting this particular family, Equation (4) more broadly points to the need for methods that operate under general confounding-shift settings. The method of Prashant et al. [62], for example, assumes overlapping confounding support, i.e., $\text{supp}(\mathbb{P}(U|\mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U|\mathcal{E}_{tr}))$, and discrete-valued proxy variables—assumptions that our analysis suggests can be relaxed.

The results in Figure 2 illustrate that, even in a simple setting without confounding overlap and with a continuous-valued proxy, a linear regression model can recover the correct causal relationship between observed covariates and the target across environments once it is given environment-specific information. This example motivates the broader need for OOD generalization methods that remain effective beyond the restrictive assumptions above—in particular, when confounding support is limited and the confounder is continuous-valued.

Minimizing residual: The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ quantifies how much additional information $\phi(\mathbf{X})$ carries about the true label Y beyond what the prediction $\hat{Y}$ already contains. It can be written as $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ relates to the expected cross-entropy loss via $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29], where $H(Y | \hat{Y})$ —the *refinement error*—measures the model’s ability to distinguish between classes. Minimizing the residual therefore admits two complementary and generally applicable strategies. First, one can directly minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ by making $\phi(\mathbf{X})$ conditionally independent of the target Y given $\hat{Y}$, i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can minimize the refinement error $H(Y | \hat{Y})$, for example through training schemes that reduce refinement error for accurate predictions before addressing calibration error [10].

4.3 Impact of additional informative variables

In § 4.2, we presented the optimal strategies for generalization when the confounder is unobserved. In many practical settings, one has access to proxies for the hidden confounding variable [4, 85, 62], which can be leveraged to substitute for it. We now study, in general terms, the impact that adding such non-causal but informative covariates has on predictive information.

Definition 4.3 (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e., neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

Since $U \rightarrow Y$, any covariate informative for U is also informative for Y , and vice versa. From the predictive information decomposition in (1), maximizing $I(Y; \hat{Y})$ requires minimizing the concept shift $I(Y; E | \phi(\mathbf{X}))$. Based on their empirical analysis, Liu et al. [50] suggest that collecting additional covariates $\mathbf{X}_I$ that make $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ more stable across environments—i.e., reducing concept shift—improves OOD test accuracy. We provide a general theoretical account of this effect, showing that using more informative variables helps maximize predictive information by increasing or decreasing certain terms in (1).

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \quad & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y | E) > I(\phi_1(\mathbf{X}); Y | E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) \quad & I(Y; E | \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E | \phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y) \end{aligned} \quad (5)$$

Proposition 4.3 shows, in full generality, that adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii) while reducing concept shift (5.iii). At the same time, it amplifies variation (5.iv), which may call for dedicated strategies to control that variation. These trade-offs hold for any informative covariate set satisfying the stated condition, independent of the particular task.

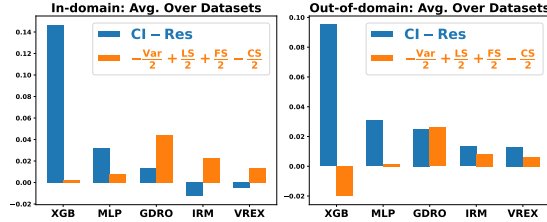
5 Experimental results

We conduct experiments on both real-world and synthetic datasets to analyze: (i) the presence of hidden confounding shift in real-world data, (ii) how the components of the decomposition in (1) affect OOD generalization under hidden confounding shifts, and (iii) the role of informative covariates in improving generalization. To span a diverse range of application domains—including public assistance, healthcare, labor, and education—we consider eight real-world tabular benchmark datasets: *Food stamps, Readmission, Income, Public coverage, Unemployment, Diabetes, Hypertension, and ASSISTments*. These datasets and their domain splits are adopted from the *TableShift* benchmark [24]. We measure model performance using ID and OOD test accuracies. To cover complementary model classes, we evaluate two ERM-based methods, XGBoost (XGB) [15] and a multi-layer perceptron (MLP); two domain generalization methods, IRM [6] and VREX [38]; and one robust-learning method, Group DRO (GDRO) [69]. We report averages over 5 random hyperparameter choices, as described in Appendix B. The effect of different random seeds is statistically insignificant for these datasets [24]. Additional setup details appear in Appendix § B, and results with hyperparameter tuning appear in Appendix C.

Hidden confounding shift: Recall that hidden confounding shifts induce observable shifts in the distributions $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$. Table 2 shows that label shift, covariate shift,

Table 2: Quantifying distribution shifts across datasets. Mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples. Statistical significance against a mean of zero is assessed via one-sample t-tests ($\alpha = 0.05$), confirming that all measures differ significantly from zero across every dataset (p-value ≈ 0).

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: Across the eight datasets, the difference *conditional informativeness* – *residual* in the plots is consistently positively correlated with the average ID and OOD test accuracy reported in the table on the right.

379 conditional covariate shift, and concept shift are all present across the real-world datasets, consistently
380 indicating hidden confounding shifts throughout. For a complementary qualitative analysis, we
381 query GPT-4o [1] to list plausible hidden confounders for several benchmark datasets [24, 57]; these
382 results appear in Appendix § E. In the *food stamps* dataset, for example, unmeasured factors such as
383 region-specific *economic policies* may influence both *household* income and *food stamp reciprocity*,
384 and analogous confounders can be identified for the other datasets.

385 **Conditional informativeness vs. accuracy:** By Proposition 4.2, under hidden confounding shift
386 the difference *conditional informativeness* – *residual* is what governs predictive information, and
387 maximizing it is essential. Results on the first six datasets (those with more than one training
388 environment [57, 24]) from Table 2 are shown in Figure 4, with dataset-wise results in Appendix C.
389 Across these datasets, the difference *conditional informativeness* – *residual* is consistently positively
390 correlated with both ID and OOD accuracy, and XGB—the best-performing model—attains the
391 highest value of this quantity. The dataset-wise results in Appendix C further reveal a broadly
392 recurring pattern: the orange bars lie close to zero because, under hidden confounding shift, the
393 sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ vanishes (Proposition 4.2).
394 The relative ID/OOD performance of the methods in Figure 1 is consistent with results previously
395 reported across the literature [57, 24].

396 **Informative covariates vs. accuracy:** We next study how including informative covariates helps
397 achieve better OOD generalization. Adopting the covariate partitioning of [57], we group all
398 available covariates into three nested subsets: *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$
399 *all covariates* (see Appendix § B), and evaluate how the terms in the decomposition (1) respond to
400 ID and OOD performance as one moves from a smaller subset to a larger one. To quantify this, we
401 introduce the *sign consistency metric*. When moving to a larger covariate set, for any term in (1) with
402 a positive coefficient, such as conditional informativeness, we count how often its value rises together
403 with accuracy; for negatively weighted terms, such as concept shift, we count how often its value falls
404 as accuracy improves. The metric is the fraction of observations in which a term’s change aligns with
405 its *beneficial* direction, thereby capturing how reliably that term contributes to better generalization
406 across settings. The formal definition is given in Appendix § B. Table 3 shows that both conditional
407 informativeness and concept shift exhibit high sign consistency with accuracy gains. While reducing
408 concept shift is broadly important, XGB additionally boosts conditional informativeness relative to
409 the other methods, yielding further performance improvements. Dataset-specific results appear in
410 Appendix § C.

Table 3: Across all methods, the sign consistency value is uniformly high for concept shift (CS). By Proposition 4.2, however, it is the sign consistency of conditional informativeness (CI) that is decisive for generalization, and XGB excels at this. Res: residual, Var: variation, FS: feature shift, C: causal, AC: arguably causal, A: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

Mean Squared Error

Cond. Informativeness & Variation

Feature Shift & Concept Shift

Legend: Linear Regression (red line with triangles), XGB (blue line with triangles), GDRO (orange line with triangles), IRM (black line with triangles), VREX (green line with circles), Cond. Informativeness (dotted line), Variation (solid line), Feature Shift (dashed line), Concept Shift (dash-dot line).

Figure 5: Adding more proxy variables $\mathbf{X}_I$ of U that are informative for Y consistently reduces MSE and increases conditional informativeness and feature shift while reducing concept shift.

In real-world datasets, it is often difficult to verify whether *all available covariates* in fact include *every relevant or sufficiently informative* variable. To test our hypothesis under fully controlled conditions and thereby probe its generality, we conduct experiments on synthetic data with a known causal structure: $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ denotes informative covariates caused by the hidden confounder U . Consistent with the theory, including $\mathbf{X}_I$ improves conditional informativeness and feature shift while reducing concept shift, as shown in Figure 5. The variation component remains at zero throughout, owing to the simplicity of this controlled setting. The synthetic data-generating process and additional results are provided in Appendix § D.

6 Discussion, limitations, and future work

In summary, we decompose the predictive information between model outputs and true outcomes to reveal, in general terms, the factors that limit OOD generalization (Equation (1)). We argue that hidden confounding shifts—reflected in covariate and outcome distribution changes across environments—are essentially unavoidable wherever data are collected across heterogeneous conditions, and that no finite set of observed covariates or proxies can fully resolve the ambiguity they induce. Because confounding entangles covariate, label, concept, and feature shifts (inequalities in (2), (3)), treating these shifts in isolation is generally ineffective. We show that learning environment-specific input–output mappings is more effective than enforcing a single invariant predictor (Equation (4)), a conclusion that holds across the tasks and model classes we study. By underscoring the inevitability of hidden confounding and the need to confront it directly, our work lays a foundation for more robust OOD methods with broad applicability. Promising directions ahead include: (i) developing algorithms that remain effective under limited confounder overlap, (ii) quantifying the cost–accuracy trade-off of acquiring informative covariates, (iii) handling entangled shifts without relying on untestable proxy assumptions, and (iv) modeling the ambiguity introduced by unobserved confounders to inspire new, broadly OOD-robust paradigms.

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713 Appendix

714 A Proofs of propositions

715 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 716 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 717 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

718 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 719 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

720 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
 721 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
 722 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
 723 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

724 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

725 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

726 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the
 727 expansion instead of $\phi(\mathbf{X})$. $\square$

728 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 729 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 730 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

731 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
 732 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

733 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 734 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

735 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 736 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 737 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

738 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

739 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

740 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

741 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 742 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 743 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

744 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 745 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

746 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

747 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

748 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

749 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 750 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

751 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 752 $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

753 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
754 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
755 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
756 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

757 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
758 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
759 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
760 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
761 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

762 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

763 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

764 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
765 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
766 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
767 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
768 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

769 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
770 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
771 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
772 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
773 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

774 B Experimental setup

775 In this section, we detail our experimental setup. The impact of random seeds on the results is
776 statistically insignificant across many settings [24]. For the results in the main paper, we report the
777 mean and standard deviation across five random hyperparameter values (sampled from their respective
778 ranges in Table B1) for each dataset–method combination. To estimate the mutual information terms
779 on both ID and OOD test data in a model-agnostic way, we use the nonparametric entropy estimation
780 toolbox [37, 77, 78]. Following [24], we evaluate mutual information using both in-distribution (ID)
781 test data (from training domains) and out-of-distribution (OOD) test data (from test domains).

782 For consistency, we draw 20,000 random samples
783 from each of the ID and OOD test sets,
784 though some datasets contain fewer than 20,000
785 samples. Across our experiments, sample size
786 has a relatively minor effect on the evaluated mutual
787 information terms. For instance, as shown
788 in Figure B1, the terms in Equation (1) com-
789 puted for the representations of the XGB model
790 on the *income* dataset remain stable as sample
791 size varies. Features $\phi(\mathbf{X})$ are extracted from
792 the layer preceding the classification head; for
793 XGBoost, we combine the model’s output mar-
794 gins and SHAP values. All experiments were conducted on a single NVIDIA RTX A6000 GPU.

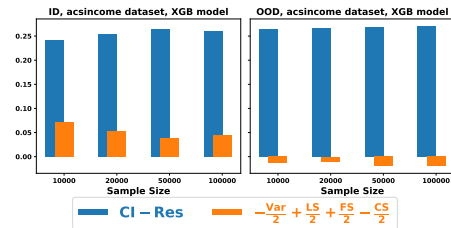


Figure B1: The evaluated terms of predictive information remain relatively consistent across a wide range of sample sizes, illustrated here on the *income* dataset.

795 In the experiments studying how informative covariates affect predictive information, we fol-
796 low the categorization introduced by Naei and Hardt [57], partitioning covariates into *causal*,
797 *arguably causal*, and *all covariates*. Causal covariates have a clearly established influence on
798 the target, with well-supported directionality and unlikely reverse causation; strict reliance on
799 them, however, risks omitting relevant parents due to gaps in domain knowledge. Arguably causal
800 covariates have uncertain causal relationships, meeting at least one of the following: (1) being
801 known causal, (2) having plausible but potentially bidirectional influence, or (3) having a likely
802 but unconfirmed causal effect. These groups approximate the true causal parents given available

Table B1: Hyperparameters and the ranges of values explored for each method in this paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

803 knowledge, though the relationships may be confounded. Some datasets additionally contain *anti-*
804 *causal* covariates, which the target likely affects rather than the reverse. The *all covariates* set
805 includes every observed variable regardless of causal status. This nested categorization is gen-
806 eral enough to apply uniformly across the diverse datasets we consider. The sign consistency
807 metric introduced in the main paper § 5 is defined as follows. For each $(m, \sigma_m) \in M$ where
808 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
809 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

Where P is the set of covariate set pairs (s, t) with $t \supset s$. For example, s can be causal covariates and t can be arguably causal covariates. There is a factor of 2 in $2P$ that accounts for both train and test data. $\mu_s^{\text{tr}/\text{te}}(m)$ is the mean of measure m for covariate set s in training/testing data. $\mathbb{I}(\cdot)$ is the indicator function. $|P| = 3$ for covariate settings $\{\text{causal}, \text{arguably causal}, \text{all}\}$. To get final sign consistency metric value for each m , we average across datasets: $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$ where $|D|$ is the number of datasets.

C Additional results on real-world datasets

In this section, we present additional results on real-world datasets. Figures C2, C3 compare the terms *conditional informativeness – residual* and *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2* for each dataset (their aggregation appears in the main paper, Figure 4 Left). Two takeaways emerge consistently across datasets. First, by Proposition 4.1, the sum of the blue and orange terms is positively correlated with overall model performance. Second, owing to the hidden confounding shift pervasive in real-world data, the difference *conditional informativeness – residual* contributes more to the overall predictive information than *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2* does. Together, these observations broadly motivate maximizing conditional informativeness for better OOD generalization under hidden confounding shift.

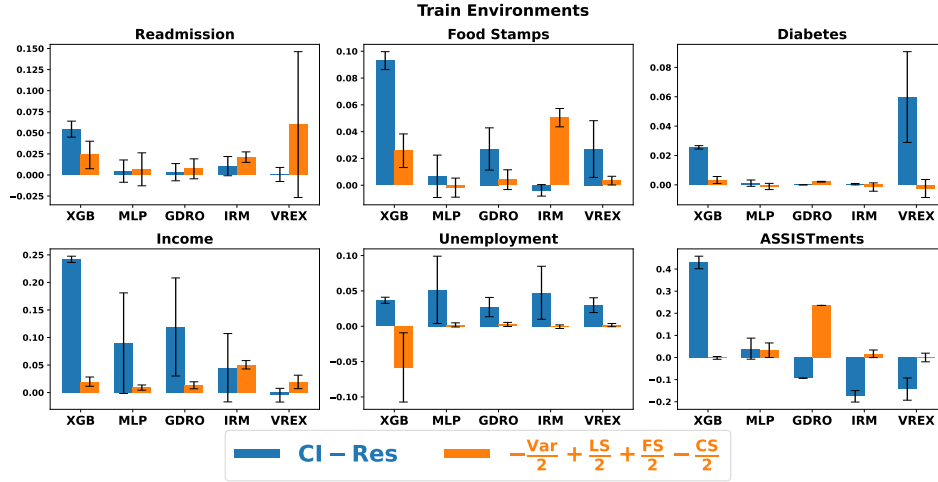


Figure C2: Per-dataset results on ID data, showing the same trend across all datasets.

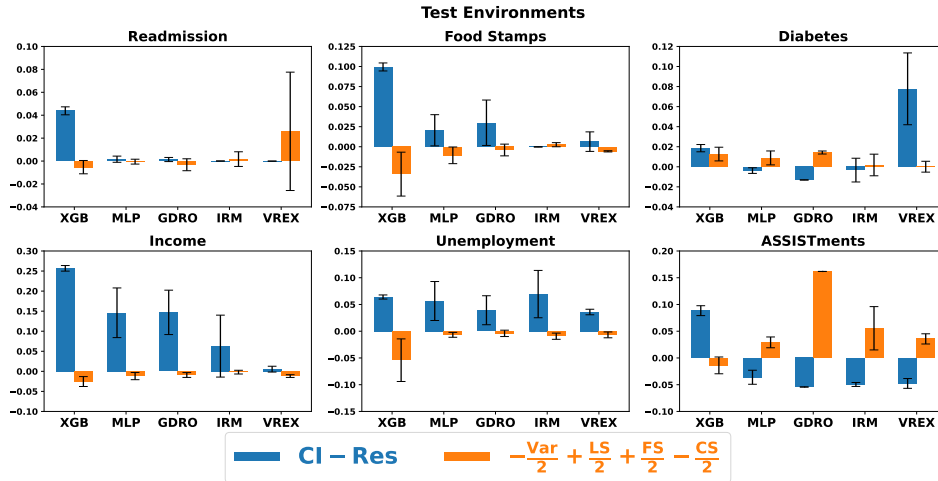


Figure C3: Per-dataset results on OOD data, showing the same trend across all datasets.

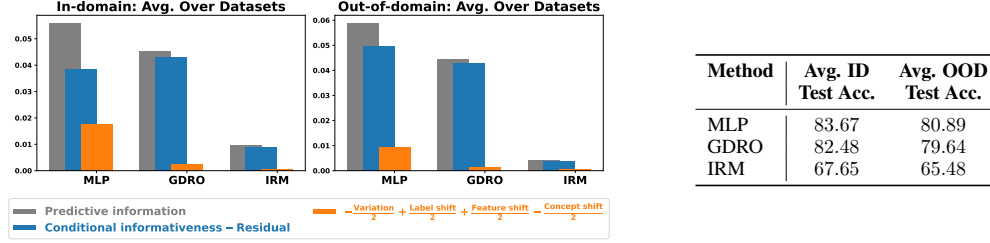


Figure C4: With tuned hyperparameters, the difference *conditional informativeness* – *residual* in the plots remains positively correlated with the average ID and OOD test accuracy over the five datasets in the table on the right. As before, this difference contributes more to the overall predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ does.

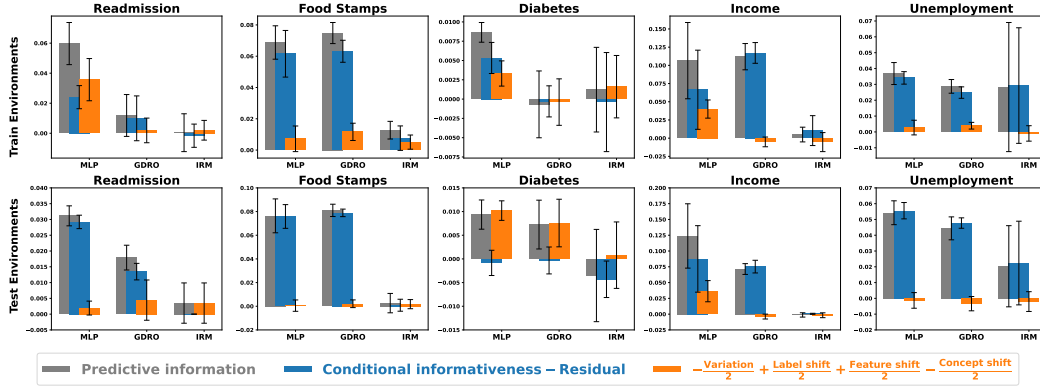


Figure C5: Per-dataset comparison of the terms in the predictive information decomposition, with tuned hyperparameters.

Table C2: Comparison of ID and OOD test accuracies of the models across datasets.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05		
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00		
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00		

Figures C6- C15 further compare the various terms of the information decomposition in (1) for every method and dataset. Across these results, the values in the second column (*variation*) and third column (*feature shift*) are similar in magnitude, so by Equation 1 their difference is close to zero and contributes little to the overall predictive decomposition. The same holds for the fourth column (*label shift*) and fifth column (*concept shift*), whose difference, again by Equation 1, is likewise near zero. Consequently, in nearly all settings the difference *conditional informativeness* – *residual* is the dominant contributor to the predictive information.

Hyperparameter tuning: We also perform hyperparameter tuning. We run a grid search over hyperparameter values and select the best configuration based on ID validation accuracy (not OOD validation accuracy). In total we run 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from a grid). To obtain mean and standard deviation for the accuracies and mutual information terms, we repeat each experiment over 5 random seeds, even though the effect of random

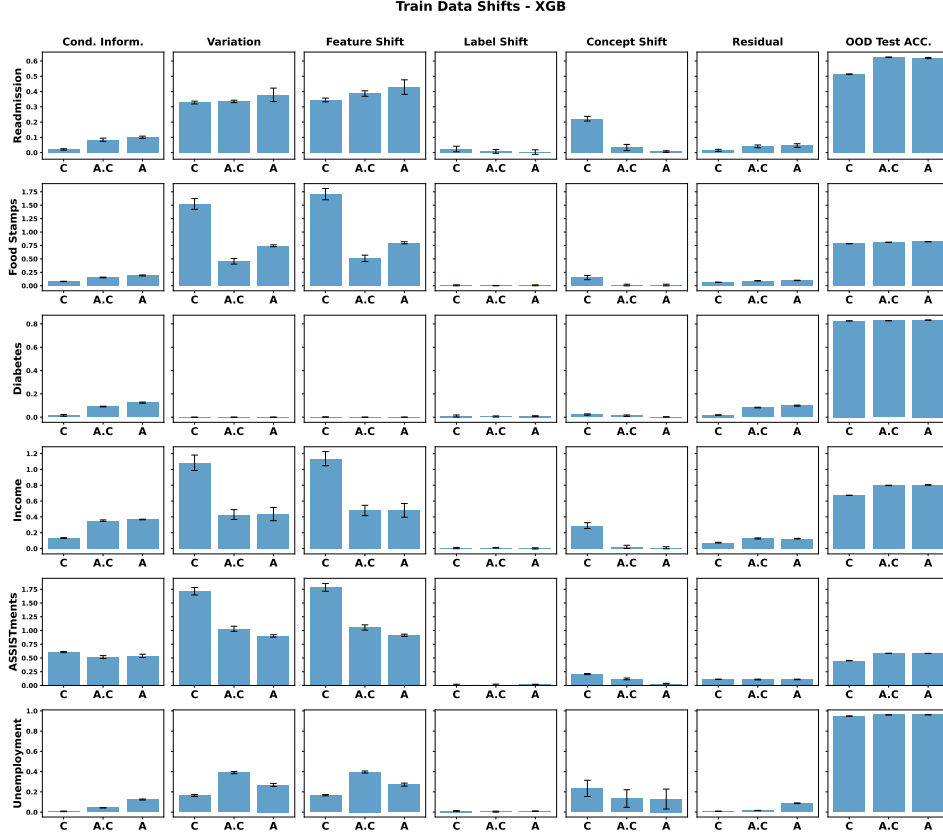


Figure C6: Per-feature-group decomposition of the information metrics on ID (training-domain) data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

seeds is insignificant in many cases. The results, shown in Figure C4, reproduce the same picture as before: by Proposition 4.1, the sum of the blue and orange terms is positively correlated with overall model performance, and, owing to the hidden confounding shift pervasive in real-world data, the difference *conditional informativeness* – *residual* contributes more to the overall predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ does. The dataset-specific results in Figure C5 and Table C2 exhibit the same trends, confirming that these findings are not specific to any single dataset.

Potential limitations of the predictive information decomposition analysis: Although predictive information $I(Y; \hat{Y})$ measures the statistical dependence between predictions and labels, it does not by itself guarantee accuracy. For example, a binary classifier that systematically predicts the *wrong* label (e.g., $\hat{Y} = 1$ when $Y = 0$) attains the maximal $I(Y; \hat{Y}) = 1$ bit through perfect anti-correlation, yet yields 0% accuracy, because mutual information quantifies *reduction in uncertainty* rather than correctness. The other decomposition terms offer additional insight: conditional informativeness $I(\phi(X); Y|E)$ may mask harmful variation $I(\phi(X); E|Y)$, while feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. Even so, as our experiments demonstrate, the decomposition remains broadly valuable for analyzing reasonably performant models: it reveals whether predictive information stems from generalizable patterns or environment-specific artifacts, enabling targeted improvements while preserving interpretability.

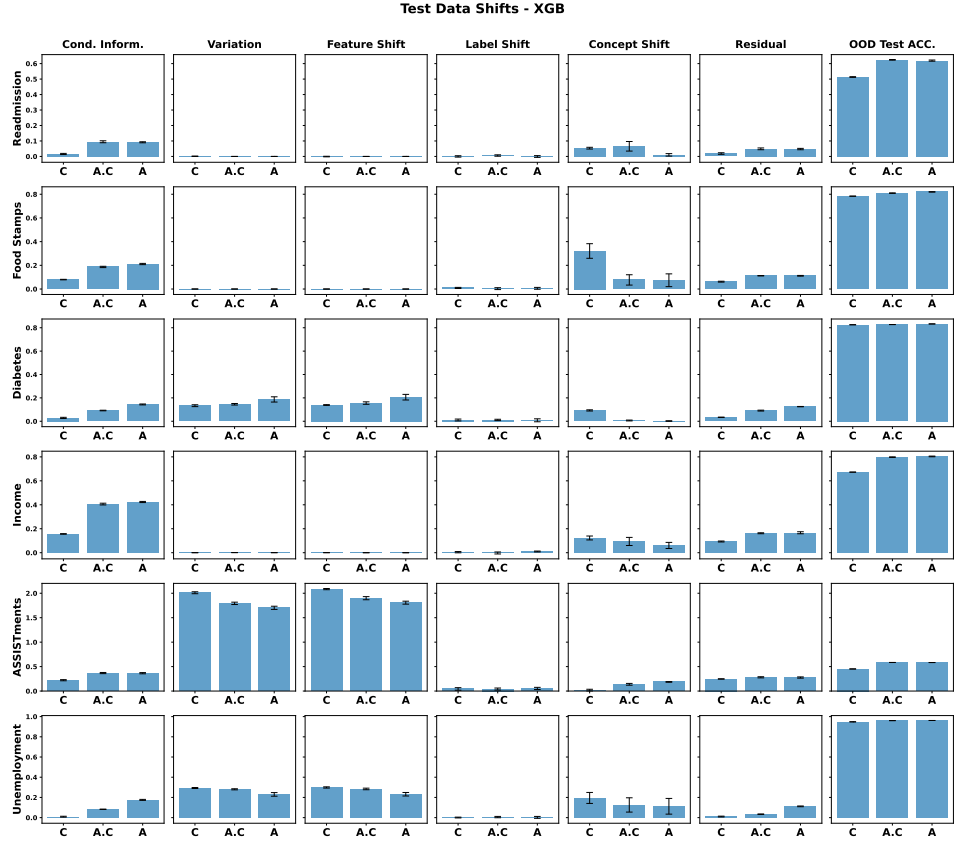


Figure C7: Per-feature-group decomposition of the information metrics on OOD (test-domain) data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

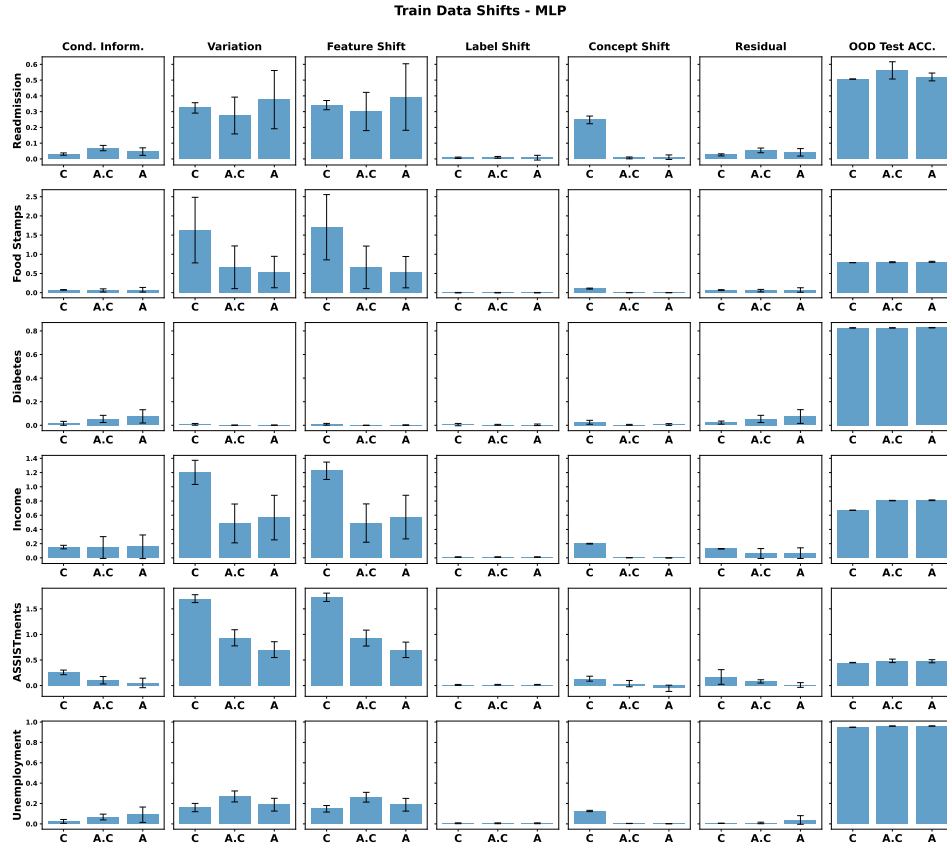


Figure C8: Per-feature-group decomposition of the information metrics on ID (training-domain) data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

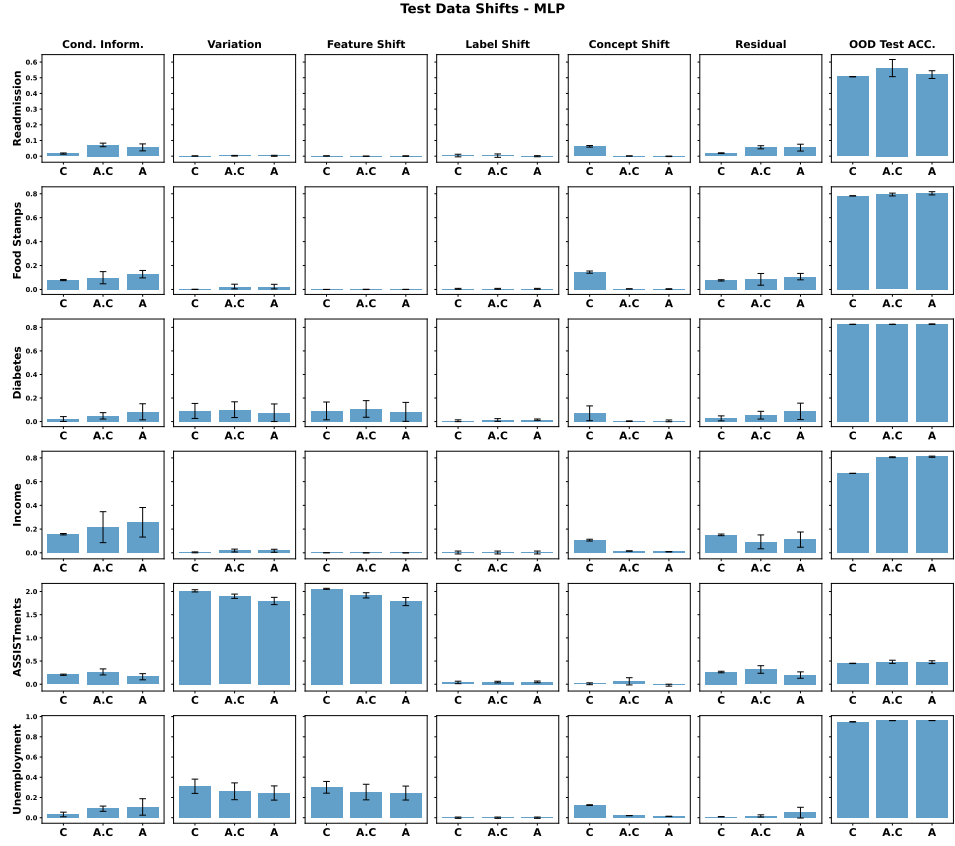


Figure C9: Per-feature-group decomposition of the information metrics on OOD (test-domain) data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

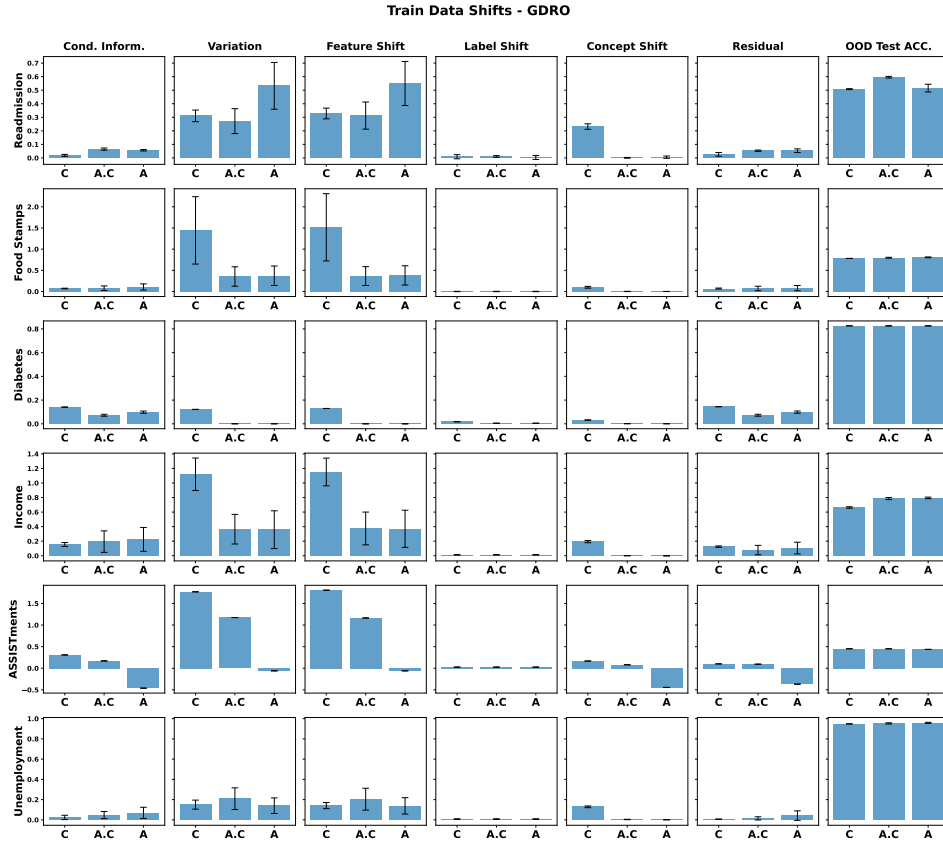


Figure C10: Per-feature-group decomposition of the information metrics on ID (training-domain) data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

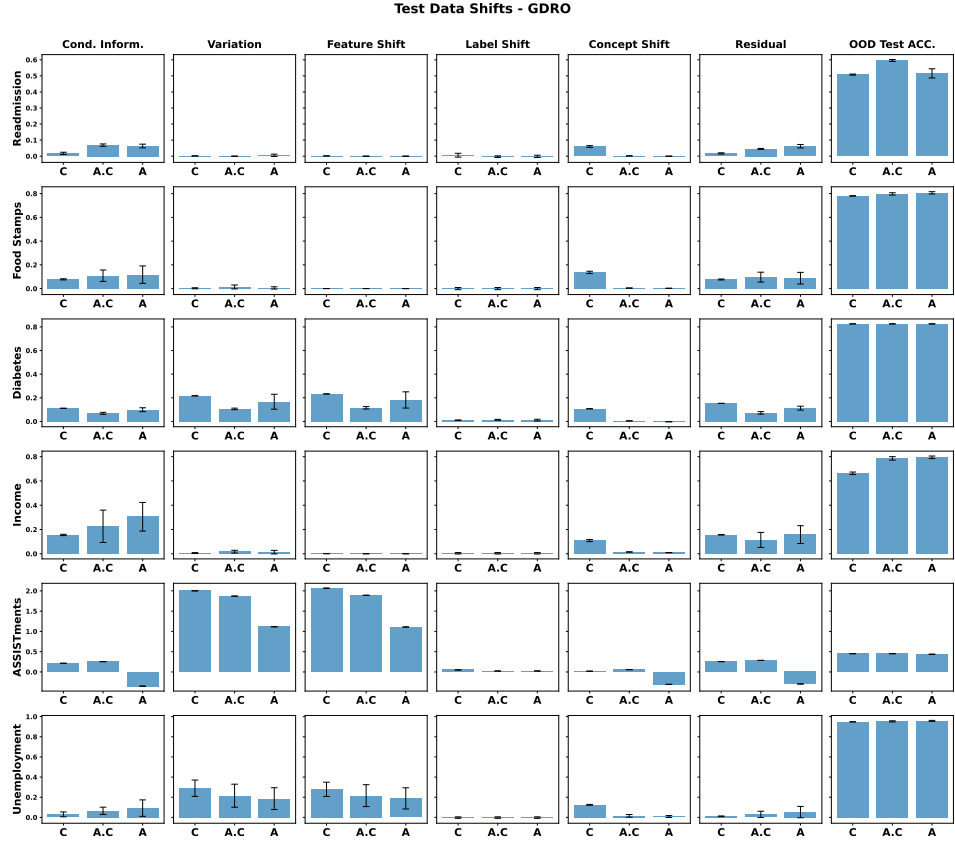


Figure C11: Per-feature-group decomposition of the information metrics on OOD (test-domain) data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

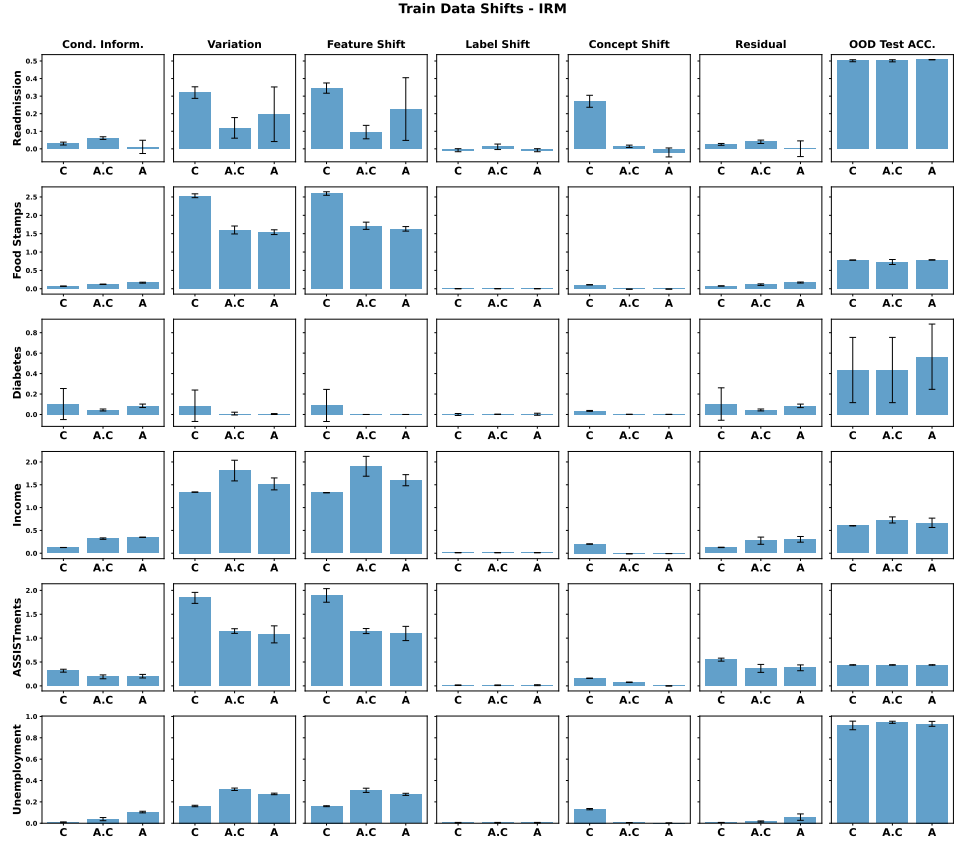


Figure C12: Per-feature-group decomposition of the information metrics on ID (training-domain) data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

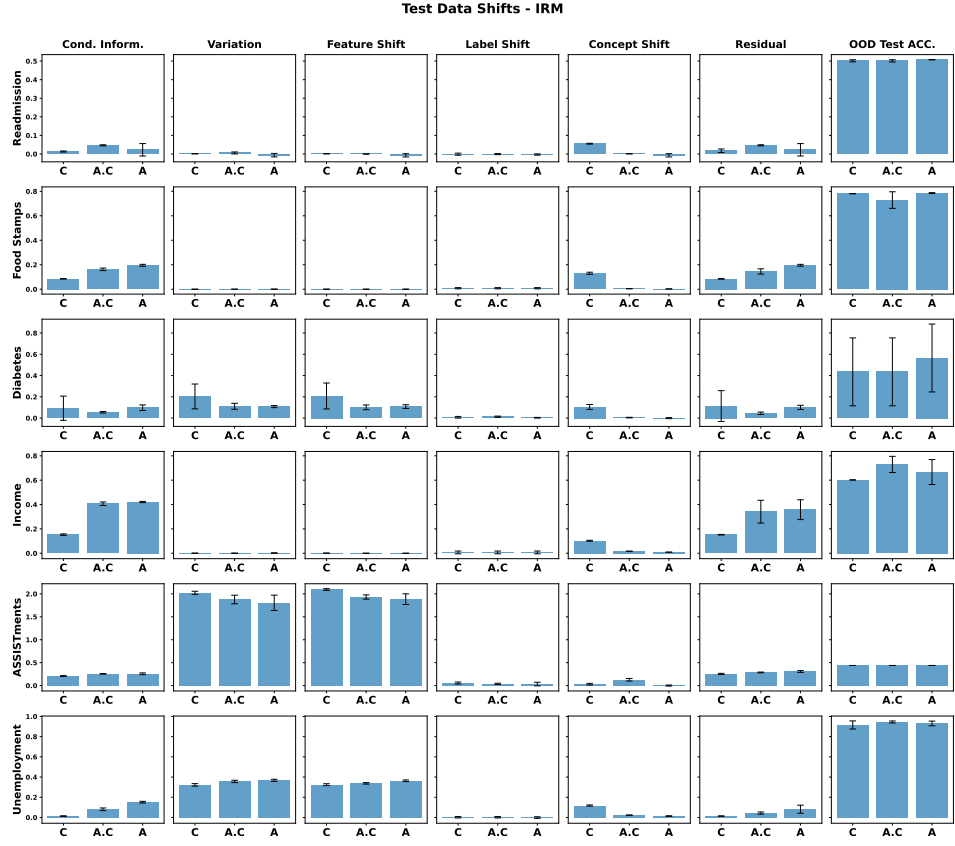


Figure C13: Per-feature-group decomposition of the information metrics on OOD (test-domain) data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

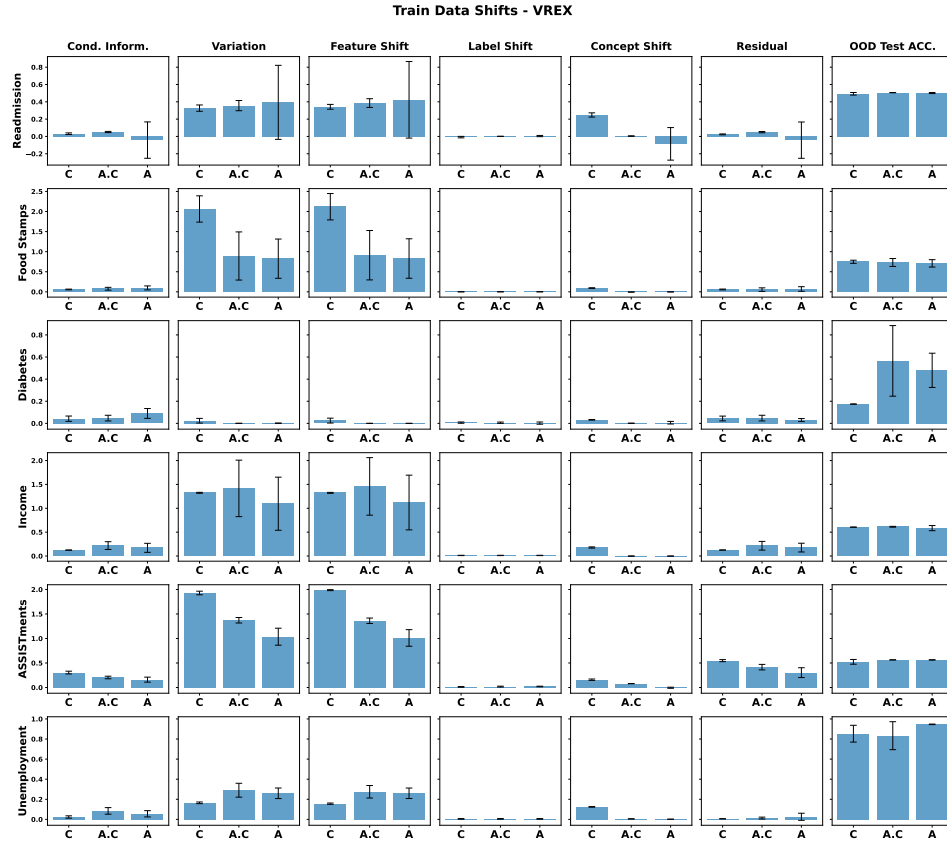


Figure C14: Per-feature-group decomposition of the information metrics on ID (training-domain) data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

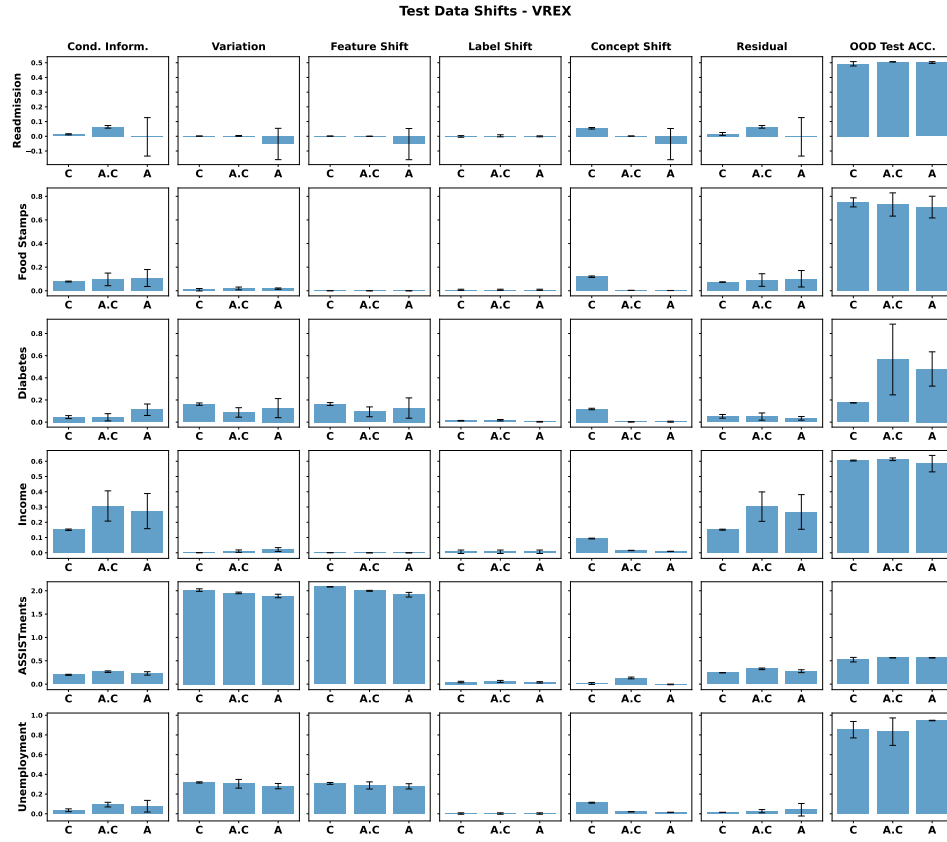


Figure C15: Per-feature-group decomposition of the information metrics on OOD (test-domain) data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

D Results on synthetic datasets

We conduct a broad suite of experiments on synthetic datasets to investigate: (i) the impact of different informativeness criteria, (ii) how the level of confounder overlap across environments affects model performance, and (iii) how the number of causal versus anti-causal variables influences performance. Using synthetic data lets us vary each of these factors independently and thereby probe the generality of our findings under controlled conditions. The data-generating process follows the structural equations in (12), with the causal structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is an unobserved confounder, X an observed covariate, Y the target, and $\mathbf{X}_I$ a set of additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

We explore a range of functional forms for f_X , f_i , and f_Y , with the corresponding results reported below. Domain shifts are induced by systematically varying the environment-specific mean parameter μ_u^e of the confounder distribution, which lets us examine how varying degrees of distributional shift affect model performance across environments.

Informativeness vs. accuracy: We begin with linear structural equations $f_X = 0.3U$, $f_i = 0.1U$, and $f_Y = X - 2U$, with $\mu_u^e \in \{-2, 2\}$ for the training environments and $\mu_u^e \in \{0, 4\}$ for the test environments. For the results in Figure 5 of the main paper, we consider one hidden confounding variable with $|\mathbf{X}_I| = 20$ informative covariates. As we increase the number of informative covariates from 0 to 20, we consistently observe (i) reduced mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, and (iv) decreased concept shift. To probe the generality of this behavior, we vary the informativeness criterion in two further ways. First, we analyze the case of one hidden confounder and a single informative covariate ($|\mathbf{X}_I| = 1$) across noise levels $\sigma_i \in \{0, 0.1, \dots, 2.0\}$. Figure D16 shows that as noise decreases from 2.0 to 0.0, the same four effects recur: (i) reduced mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, and (iv) decreased concept shift, with the variation term remaining zero throughout, as in the other synthetic experiments. Finally, we study a setting with 20 hidden confounders and 20 corresponding informative covariates, using the modified equations $f_X = 0.3U$, $f_i = 0.2U$, and $f_Y = X - 2U$. Figure D17 shows that introducing informative covariates for the hidden confounders again produces (i) reduced mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, and (iv) decreased concept shift. Across all three variants, the same pattern holds, demonstrating that a variety of informativeness criteria help achieve OOD generalization.

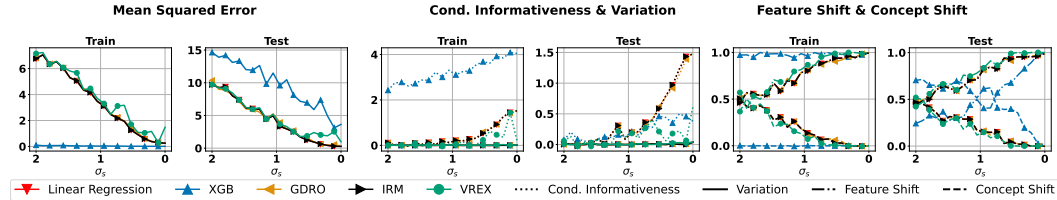


Figure D16: Adding a low-noise proxy variable consistently reduces MSE and increases conditional informativeness and feature shift while reducing concept shift.

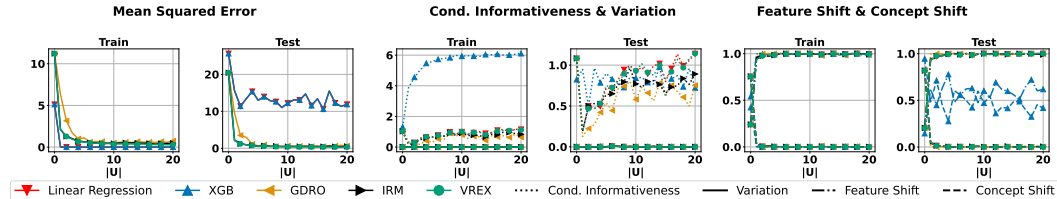


Figure D17: Adding more proxy variables $\mathbf{X}_I$ for a set of hidden confounding variables $\mathbf{U}$ that are informative for Y consistently reduces MSE and increases conditional informativeness and feature shift while reducing concept shift.

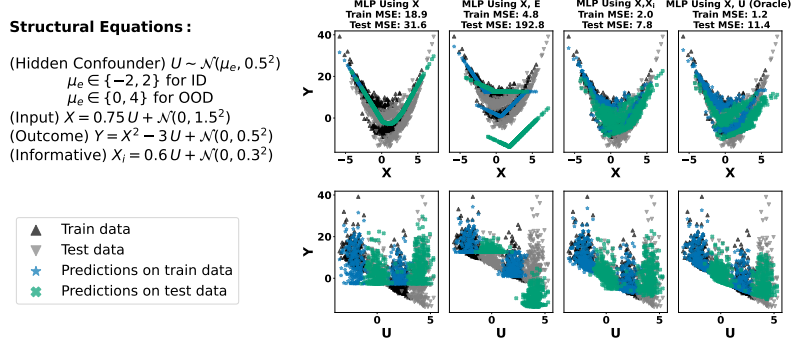


Figure D18: Performance of MLP in the low-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

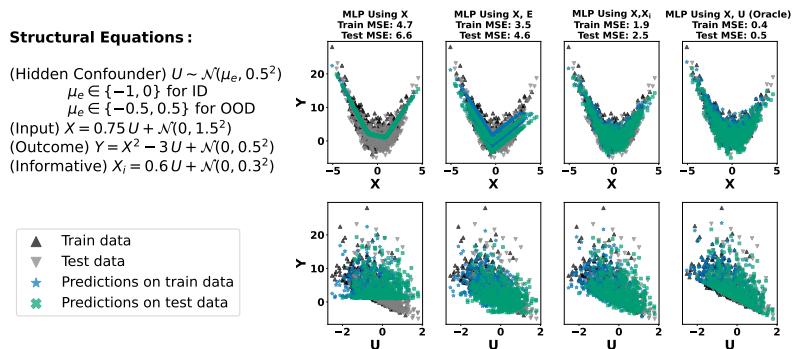


Figure D19: Performance of MLP in the high-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

Overlapping confounder support vs. accuracy: As discussed in the main paper, recent work by Prashant et al. [62] proposed an OOD generalization method for hidden confounding shift that assumes the test confounder support remains within the training support. Having shown that correct relationships can be learned in synthetic linear data whenever the observed covariates carry sufficient information about the hidden confounders, we now extend the analysis to nonlinear data using MLP and XGB models under both low and high confounding-support conditions (Figures D18–D21), thereby testing the argument well beyond the linear case. In the low-overlap setting (characterized by distant μ_e between ID and OOD data), models relying solely on X learn an inadequate global function that fails to distinguish environments. Performance improves once environment-specific summary statistics of the observed covariates are incorporated, capturing environment-specific relationships, and the largest gains arise only when additional informative covariates are leveraged—again underscoring their importance. For completeness, we include oracle-model results and U – Y scatter plots to elucidate the learned input–output mappings. As expected, these experiments confirm stronger performance under high confounding overlap; more importantly, they reveal encouraging results in the low-overlap regime—a previously understudied scenario that challenges the common-support assumption prevailing in the literature and suggests that the approach generalizes beyond it.

Number of causal vs. anti-causal variables: Recent benchmarks [24] show that baseline methods (e.g., XGB, MLP) consistently match or outperform OOD generalization techniques (e.g., GDRO, IRM, VREX) across real-world datasets, coinciding with the prevalent causal structure $\mathbf{X} \rightarrow Y$ in tabular data. To test whether this baseline superiority stems from the $\mathbf{X} \rightarrow Y$ structure, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (e.g., $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$), sweeping the full spectrum from purely anti-causal to purely causal covariate mixes. The results (Figure D22) show that baselines remain resilient even under the causal structure $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$, though no single method dominates universally (Figures D23, D24); baselines nonetheless achieve superior PMA-OOD scores [24] (fraction of maximum OOD accuracy). To ensure the comparison spans a representative cross-section of model classes, we

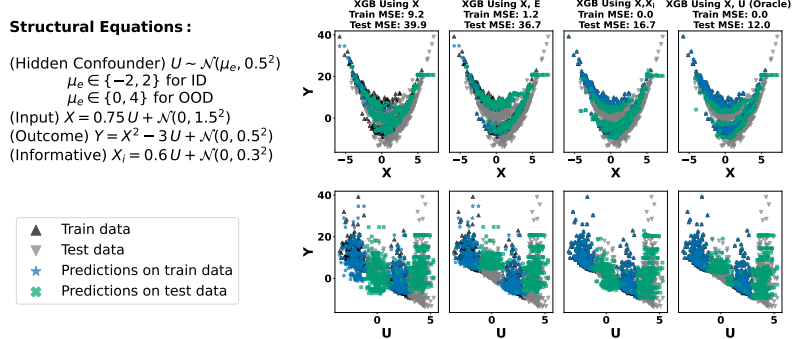


Figure D20: Performance of XGB in the low-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

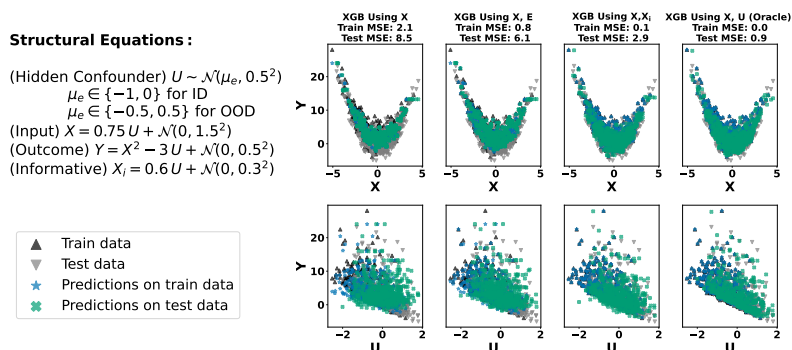
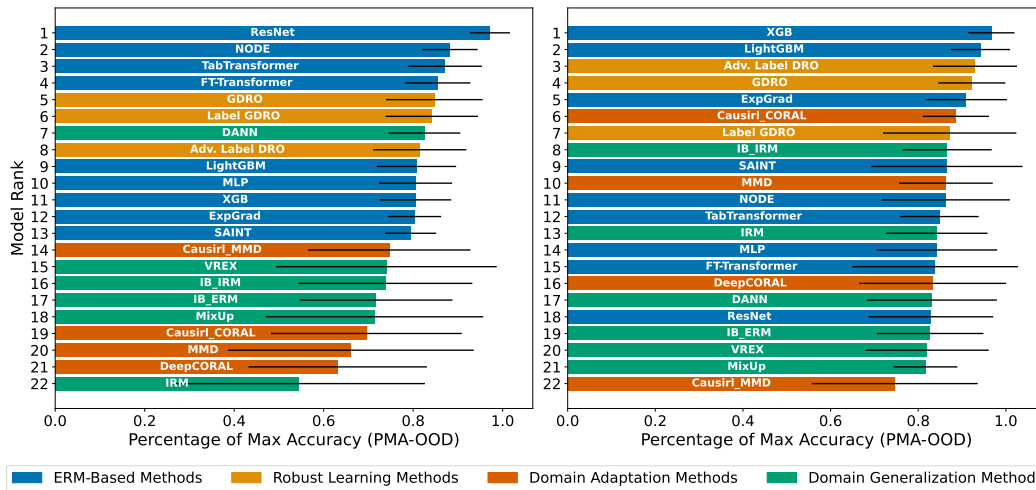


Figure D21: Performance of XGB in the high-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle).

912 evaluate: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76],
913 TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods
914 (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-
915 REx [38], Domain Mixup [90, 91]); and (3) robust optimization methods (DRO [42], GroupDRO [69],
916 Label DRO, Adversarial Label DRO [94]).

Figure 1 displays OOD Test Accuracy for various methods across different values of ρ (0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0). The methods are categorized into four groups: Domain Generalization Methods (green), Domain Adaptation Methods (orange), Robust Learning Methods (yellow), and ERM-Based Methods (blue). The x-axis for each chart lists 15 methods: DGM, DGM+, DGM+, DGM+, DGM+, DGM+, DGM+, DGM+, DGM+, DGM+, DGM+, DGM+, DGM+, DGM+, DGM+. The y-axis represents OOD Test Accuracy from 0.0 to 1.0. As ρ increases, the performance of most methods decreases, with ERM-Based Methods generally showing the highest accuracy and Domain Generalization Methods showing the lowest.

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ($\rho = \frac{|\mathbf{X}_{cl}|}{|\mathbf{X}_a|}$)
 Left: Data With Linear Equations, Right: Data With Nonlinear Equations



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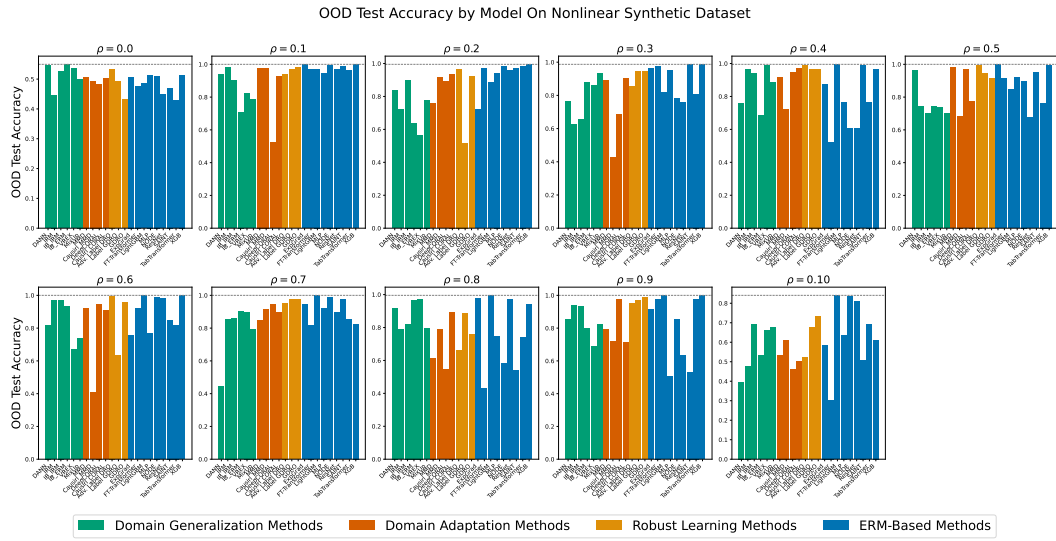


Figure D24: Per-model OOD test accuracy in the nonlinear setting across the full range of ρ values.

917 E Qualitative analysis of hidden confounding shift in real-world datasets

918 In this section, we qualitatively examine the presence of hidden confounding shifts across a wide range
 919 of real-world datasets. Tables E3-E18 list plausible hidden confounders influencing various observed
 920 covariates in many real-world datasets [57]; for additional details on these datasets, we refer to
 921 [24, 57]. We use GPT4o [1] for this task and note that the enumerated set of unobserved confounders
 922 is not exhaustive. The breadth of datasets covered—spanning public assistance, healthcare, education,
 923 and beyond—reinforces the argument that hidden confounding shifts arise and matter across diverse
 924 domains rather than in isolated cases.

925 **Prompt:** We use the prompt below to query GPT4o to get possible hidden confounders.

926 For a target variable, you will be given lists of causal, arguably
 927 causal, anti-causal, and spurious covariates. Also, you will be provided
 928 with environment variables such that different datasets are collected in
 929 environments induced by these environment variables. Provide possible
 930 hidden confounders that cause any of the target, causal, arguably causal,
 931 spurious, and anti-causal covariates. That is, the hidden confounders
 932 should influence the distribution of their children in different domains.
 933 Ensure hidden confounders are not a part of the given variables, and your
 934 explanation should include how hidden confounders influence their children
 935 in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.